

0. Commenting & Printing

```
# This is a comment  
print()
```

Operating System	Shortcut
Mac OS X	CMD + /
Windows	CTRL + /

1. Data Types

Data Type	Description
int	Integer numbers
float	Decimal numbers
str	Sequence of characters bounded by single or double quotes
bool	True or False
None	Nothing – represents absence of a value

type() checks a value's data type

Function	Description
int()	Casts value as an int
float()	Casts value as a float
str()	Casts value as a str

2. Variables

Declare a variable with **=**

Variable Naming:

1. Can't start with a number or special character
2. No spaces allowed
3. Use only letters, numbers, and underscores _
4. Be descriptive with names

3. Operators

Arithmetic Operators

Operator	Description
+	Addition
-	Subtraction

*	multiplication
/	division
**	exponent

Comparison Operators

Operator	Description
==	Equal in value
!=	Not equal in value
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

4. Data Structures

Property	list	tuple	dict	set
Ordered?	Yes	Yes	Yes	No
Change items?	Yes	No	Yes	Yes
Duplicate items?	Yes	Yes	No duplicate keys	No

5. Lists

```
mylist = [1, 2, 3]
```

Working with lists

Task	Syntax
Find list length	len(mylist)
Access list items with indices	mylist[0]
Add item to end of list	mylist.append(item)
Add item to specific index	mylist.insert(index, item)
Remove item from list by value	mylist.remove(value)
Remove item from list by index	mylist.pop(index)
Empty a list	mylist.clear()

6. Tuples

```
mytuple = (1, 2, 3)
```

Working with tuples

Task	Syntax
Find tuple length	<code>len(mytuple)</code>
Access tuple items with indices	<code>mytuple[0]</code>
Remove/add/change item from tuple	ERROR: Can't change tuple

7. Dictionaries

```
mydict = {"Start": 1,  
          "Middle": 2,  
          "End": 3}
```

Working with dictionaries

Task	Syntax
Find number of mappings	<code>len(mydict)</code>
Use keys to access dict values	<code>mydict["Start"]</code>

8. Sets

```
myset = {1, 2, 3}
```

Working with sets

Task	Syntax
Find set length	<code>len(myset)</code>
Access items with indices	ERROR: Sets are unordered so cannot use indices
Add duplicate items	No error – sets will just ignore duplicates
Empty a list	<code>mylist.clear()</code>

9. For Loops

```
for item in data_structure:  
    # do something to each item
```

10. If Statements

```
if condition_1:  
    # do this if condition_1 True  
elif condition_2:  
    # do this if condition_1  
    False & condition_2 True  
else:  
    # do this if none of above  
    True
```

Operator	Description
and	Returns True if both statements true
or	Returns True if at least one of the statements is true

Loop Interruptions

Syntax	Description
break	Exists a loop
continue	Skips to the next iteration of a loop

11. Functions

```
def function_name(parameters):  
    """ Description here  
    Parameters  
    -----  
    parameters: description of  
    parameters  
  
    Return  
    -----  
    what is returned  
  
    """  
    # do task here  
    return value
```

Accessing a Function's Docstring

Syntax	Description
<code>function_name?</code>	Returns function's docstring